

dehip: Smoke Run Findings

HIP cascade harness (0.6B and 4B), issue 16

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Smoke Run Findings (0.6B cascade, issue # 16)

First end-to-end run of the whole pipeline on real instruments: a 50-pair FineWeb smoke corpus, Qwen3-4B-Instruct drafts, the released Qwen3-0.6B HIP paraphraser for the rewrite, the real nvidia/llama-embed-nemotron-8b embedder for MMD, GPT-5.4-mini for JMQ, and Pangram for the external detector. Date: 2026-08-07.

The 0.6B is a deliberately weak smoke paraphraser. It was chosen to validate the pipeline and the hip-run boundary cheaply, not to produce quality. The quality verdict waits for the 4B run.

The pipeline ran end to end

build-corpus (50) -> self-check PASS -> generate (50 drafts)
-> rewrite (50, real hip-run 0.6B) -> score (3 metrics) -> report -> detect

Every stage ran against real models and real APIs. The self-check gate passed first (human-vs-human MMD 0.0003), so the metrics are trustworthy.

Scorecard

Criterion	Result	Detail
SC-001 self-check	PASS	human-vs-human MMD 0.000307, token-L2 0.0221
SC-003 rewrite closer to human on 2 of 3	PASS	MMD and token-L2 yes, JMQ no
SC-004 degeneration at or under 5%	FAIL	14 of 50 (28%), the weak 0.6B
SC-005 detector at least 30 points more human	FAIL (narrow)	delta +27.3 points, threshold 30
SC-006 full run under one day	PASS	a few hours end to end

The numbers (36 common pairs that completed both rounds)

Metric	Draft	Rewrite (0.6B, k2)	Direction
MMD (lower is human)	0.0132	0.0014	9x closer to human
token-L2 (lower is human)	0.0358	0.0276	closer to human
JMQ (higher is human)	0.611	0.222	worse on every dimension
Pangram mean human-prob	0.000	0.273	+27.3 points

JMQ dimensions, draft -> rewrite: clarity 0.72 -> 0.39, coherence 0.56 -> 0.22, creativity 0.56 -> 0.22, depth 0.28 -> 0.11, relevance 0.44 -> 0.33.

Pangram distribution: all 36 drafts scored 0.0 (fully AI). The rewrites were bimodal: 9 of 36 flipped to 1.0 (fully human), 26 stayed at 0.0.

The finding

Every “looks human” metric moved the right way and the one “is it good writing” metric moved the wrong way. MMD dropped 9x, token-L2 dropped, Pangram rose 27 points with a quarter of texts fully fooled. JMQ fell on every dimension. The 0.6B HIP rewrite is a humanizer and a detector-evader, and it works even at 0.6B, but it is not a prose improver. A detector-only or MMD-only evaluation would call this a win. The two-metric-family harness catches that the rewrites are more human and worse writing at the same time.

The 0.6B rewrite’s MMD (0.0014) is lower than the published 14B DFT benchmark (0.018). A tiny degeneration-prone paraphraser beats a 14B production model on the distribution metric, by making text blander and more repetitive, which is closer to the human distribution in embedding space. Its JMQ (0.22) is far below any published row. This is the clearest possible argument for reading the metrics together, never one alone.

The open question, restated

HIP’s published wins are detector-evasion numbers. The unanswered question is whether the rewrites are better prose. For the 0.6B the answer is no. The 4B run tests whether a stronger paraphraser can pull the distribution toward human and clear the detector without the JMQ collapse. Everything needed to run it now exists and is proven.

Integration bugs the real run surfaced

Every stage was unit-tested against mocked seams, so each real integration boundary hid a mismatch that only a real run could expose. Eight in total:

1. OOM: `list()` over the unbounded `FineWeb` stream. Bounded with `islice`.
2. Protocol-isinstance class-factory: `isinstance(cls, runtime_checkable_Protocol)` is true for the class itself, so the CLI’s client class was never constructed.
3. The pre-commit hook never gated ruff (no lint target). Added just `lint`.
4. The corpus manifest carried no `{pair_id, text}` texts file, so the score reader `KeyError`d on text (the `Pair` stores it as `reference_text`).
5. The embedder needs `trust_remote_code=True` (custom `llama_bidirec` architecture).
6. The `hip-run` seam was built against an assumed interface. The real `hip-run` takes `--config` only, reads `adapter_path` and `input_jsonl` and `output_parquet` as config keys, writes a

parquet, infers the base model from the adapter, and runs on CPU (no MPS). A full seam rework, plus a round-filter bug (hip-run stamps round 1 on every single-round invocation, so filtering on the cascade round skipped every row on round 2).

7. The detector assumed a pangram package with `predict()` returning `ai_likelihood`. The real SDK is `pangram-sdk`, `predict()` returns `fraction_human`, and `model="default"` is required after 2026-09-30.

The lesson: a mock of a seam encodes your assumptions about an interface, never the interface itself. The three external seams (hip-run, the embedder, the detector) each need one real-integration smoke test, because unit mocks structurally cannot catch “the interface I imagined is not the interface that exists.”

4B run (the real quality test)

Re-ran the cascade with the released Qwen3-4B HIP adapter on the same drafts. The 4B base runs on CPU through hip-run (no MPS), and it finished in a similar time to the 0.6B because early hard-trips kept the subprocess count down. 15 of 50 flagged degenerate, so SC-004 still fails. But the quality story changed sharply.

0.6B vs 4B (rewrite, both against human)

	MMD	token-L2	JMQ	JMQ drop from draft	SC-005 Pangram delta
0.6B	0.0014	0.0276	0.222	-0.39 (collapse)	+0.273 (FAIL)
4B	-0.0020	0.0239	0.457	-0.11 (mild)	+0.333 (PASS)

The 4B rewrite’s MMD is essentially zero (statistically indistinguishable from human-vs-human), so its distribution is fully human. Its JMQ is double the 0.6B’s, the quality penalty shrank from -0.39 to -0.11, and it clears the SC-005 detector bar the 0.6B missed (10 of 35 rewrites scored fully human on Pangram).

The finding, refined

The story is a scaling curve, not a yes or no. Bigger paraphraser, smaller quality penalty. 0.6B drops JMQ by 0.39, the 4B by only 0.11. The line is heading toward quality-neutral, and an 8B or 14B might cross into positive. But at 4B, every JMQ dimension still dropped, so HIP is still “more human, slightly worse writing,” never “more human, better writing.” The honest answer to the open question, at the tiers runnable here: humanization approaches quality-neutral as the model grows, but it is not yet prose improvement. That is a more interesting result than either “it works” or “it does not.” It is a trajectory, and the two-metric-family harness is what makes it visible.

4B scorecard

Criterion	Result
SC-001 self-check	PASS
SC-003 closer on 2 of 3	PASS (MMD and token-L2 yes, JMQ still down but only -0.11)

Criterion	Result
SC-004 degeneration at or under 5%	FAIL (15 of 50)
SC-005 detector at least 30 points	PASS (+0.333)
SC-006 under one day	PASS

Known follow-ups

- DONE (a502641): test isolation. The score and self-check CLI tests defaulted to the real `data/emb-cache/`, so a real run's dim-4096 vectors collided with the stub tests' dim-4 vectors. Fixed with a `DEHIP_EMB_CACHE_DIR` override and an autouse confest fixture that points every test at a tmp cache dir.
- DONE (2c3a2a0): `bounds.py` `REAL_INSTRUMENT_BOUNDS` filled from this run's self-check (MMD 0.000307, token-L2 0.0221) with generous single-run margins. `documented()` is now instrument-aware, selecting the real bounds only for the real embedder and the stub bounds otherwise. Still open: re-derive over multiple seeds (like the stub's 24) for tight bounds instead of one observation.
- SC-004/SC-005 pass and the real quality verdict need the 4B (or larger) adapter. The 4B run cleared SC-005; an 8B or 14B GPU run would test whether the JMQ penalty crosses into quality-neutral. The recipe and sizing are ready in `docs/gpu-runbook.md` (just `fetch-adapter 8B` then just `run-cascade 8B`); it needs a CUDA box, which this Mac is not.